Physical chemistry cume for Oct. 17, 2003

At Earth's distance from the Sun, the intensity of sunlight is about 1350 W/m^2 outside the atmosphere. The dominant wavelength is about 500 nm (green). To simplify the calculations, assume in what follows that *all* photons emitted by the Sun have a wavelength of 500 nm. Photons, just like heavy particles, have mechanical momentum, given by h/λ where h is Planck's constant (h=6.62608 J/s) and λ is the wavelength.

(a) What is the photon flux in $m^{-2}s^{-1}$, i.e. the number of photons which hit a 1 m^2 area perpendicular to sunlight at the Earth's distance from the Sun?
(b) What is the force arising from light pressure on a 1 m^2 absorbing surface of a spacecraft, perpendicular to the Sun's rays, at the Earth's distance from the Sun?
(c) According to Planck, the energy radiated by a black body at absolute temperature T in a narrow wavelength range $d\lambda$ is proportional to $\rho(\lambda)=\lambda^{-5}(\exp(hc/kT) - 1)^{-1}$ where $\exp(x)=e^x$, and k is the Boltzmann constant, k=1.38066E-23 J/K , and c is the speed of light, 2.9979E8 m/s. The maximum of $\rho(\lambda)$ as a function of λ is, in very good approximation, at $\lambda_{max}= hc/(5kT)$. The maximum energy density of sunlight, (outside the atmosphere) is at 470 nm. Assuming that the Sun is a perfect (black-body) radiator, which is largely true, estimate the temperature of the Sun's surface.